

Customer Requirement Specification

CRS stands for customer requirement specification. It is designed by the customer, who has/have the requirement of a software. CRS includes all the requirements for an intended service, which provides clear idea to the developer about the development process and requirements of software. For example, an online problem solving website can be used by users for multiple purpose. Such as coding practice, learning, interview preparation etc.

Introduction:

This CRS document will provide broad explanation of requirements to make a problem solving website. Such as what are the functionalities are required to do by admin, judge and users. Also, the problem solving software should be ready within 2 months.

General description:

On the website the user should have the access as participant for coding practice, competitive programming, and interview preparation. Also, there must be a support section which should be used by users to get support from each other. Moreover, admin should access with Installing and updating system, creating filling system, monitor, user administration, implement security, organizational and problems solving abilities, Helps resolve problem etc. Also, there might be judge panel, who will set the test cases, evaluate the solutions of problems with (AC, WA, TLE, MLE, CE and others)

Functional needs:

User management:

To get any service user must register and login on the website. But non registered users should have access on the home page to see the environment of the website.

Each of the user must have a profile to showcase the interests and problems solving skills.

In case password get forgotten, recovery option should be included.

Problem submission and display:

There must be also community support. The users can share their thought, challenges and intuitive idea.

Users need to have the ability to categorize, tag, use media (image and video) on the post.

A system to see the submitted problems clearly.

Problem set:

The website need to contain many more practice problem set.

Admin should have the ability to create new problem, update existing problem, delete any problem. Also judge can change the test case of problem.

Search and discovery:

It needs to have a filter section to find specific problems with selected topics. Problem difficulty should be included.

A weekly suggested problem list should be visible for solving.

Contest:

There should be the option to create competitive programming contest by admin. And user need to get the notification before the contest. Admin should have the ability to create the announcement for the contest and sending email.

Integration and Collaboration:

Commenting, rating or upvoting features to highlight useful content.

One to one messaging and group chat for collaboration.

User engagement and gamification:

User should win the awards, badges, rating or points for solving problems on contest.

Leaderboard to show top contributors.

Leaderboard to show the top performer on the contest.

Non-Functional Requirements

Performance:

The website must be fast and responsive, even with many users or large amounts of data.

Security:

Measures to protect user data, prevent unauthorized access.

Usability:

The site should be easy to navigate and understand for all users, regardless of their technical skills.

Compatibility:

The website should function correctly across different web browsers and devices (desktops, tablets, mobile phones).

Reliability:

High uptime and availability to ensure the site is accessible when users need it.

Business Goals & Strategy

Clear Purpose:

The website will contain practice problem, contest, tutorials for learning, editorial and explanation for practice problem.

Target Audience:

The website will be for school and college level students. Who are intended to learn programming as beginner.

Success Metrics:

How will you measure the website's performance and success (e.g., number of solved problems, user engagement, community growth)?